

Who measures conversion

Testability, blindness and the missing denominator in Brazil's critical-minerals doctrine

Antoine Bachelin Sena
Institut Atlantique Sud
ORCID 0009-0004-8640-3130

IAS Working Paper 2026/01
DOI 10.5281/zenodo.22838124 (version 5; concept DOI
10.5281/zenodo.22567731)
Licensed CC BY 4.0

Data extracted from Comex Stat on 6 September 2026; the inbound 2846.90.90 line and subheading 2846.90 as a whole re-extracted on 18 September 2026.

Replication data and code accompany this paper.

Version 5, 18 September 2026. Corrects the description of NCM 2846.90.10 in Section 5.2, which version 4 described as a residual heading for rare-earth compounds excluding cerium; the line is praseodymium oxide. Records the four national lines under subheading 2846.90 and their treatment. Reclassifies 2846.90.10 from nomenclature failure to missing industry, and revises the counts in the abstract, Section 6.4, Section 6.5 and Section 10 accordingly. Revises Act One in Section 8.3 to reflect the cut already made. Replaces Section 8.1, which described a bill awaiting sanction, with an account of Law 15.506/2026, Decree 13.118/2026 and Complementary Law 235/2026. In Section 7.2b, the 2025 inbound figure under 2846.90.90 is taken from the unfiltered series re-extracted on 18 September 2026 — 887,078 kg at 1.99 US\$/kg, in

place of the thresholded 874,784 kg at 1.60 — and the outbound-to-inbound comparison becomes ninefold rather than elevenfold. Data, method and results are otherwise unchanged.

Abstract

Brazil's critical-minerals doctrine names domestic processing as its goal and supplies no measure by which a government could later establish whether it occurred. This paper proposes a falsifiable test of conversion and then asks a prior question: on how many of Brazil's public trade lines can conversion be measured at all?

Applying four conditions — sufficient mass, declared grade, a reference price independent of the exporter, and a classifiable counterparty — to Comex Stat monthly data for 2016–2025, the test proves runnable on one of eight critical-mineral lines, and that line returns a commodity reading: no Brazilian line currently validates conversion. Of the seven lines on which the test cannot be run, three are blocked by nomenclature, three by the absence of a domestic industry, and one by market structure. Only the first family is corrected by a customs code.

A second route requires no licensed price series. Destination concentration, relationship duration, rotation and unit-value dispersion separate an industrial offtake from a commodity sale using public data alone, and they do so more cleanly than the price route.

The rare-earth case is examined in detail. In 2025 Brazil exported 677.9 tonnes under NCM 2846.90.90 to China for US\$ 11.96 million: 99.78 percent of the unfiltered national line, declared from a single municipality. Unit value on that destination fell 51 percent as volume rose elevenfold, and the public statistic cannot establish whether price, grade or product changed. Following the 3 September 2026 closing of Serra Verde into USA Rare Earth, that line will become an intra-group transfer price once deliveries under the offtake begin, with no residual market portion to anchor it.

Keywords: critical minerals; industrial policy; customs nomenclature; rare earths; Brazil; trade statistics; policy evaluation.

1. Introduction

On 31 August 2026, CEBRI published Philip Yang and Fernanda Cimini's account of what they call Brazil's doctrine of conversion: the proposition that the country holds seeds of power rather than power, and that the state's task over the coming decade is to conclude, coordinate and lead — to finish an unfinished state, to coordinate what is currently dispersed, and to lead the international rule-writing that determines what Brazilian assets are worth (Yang and Cimini 2026). Two days later, on 2 September, the Senate approved the bill instituting the National Policy for Critical and Strategic Minerals. The following day, USA Rare Earth completed its acquisition of Serra Verde, the operator of Brazil's only commercial rare-earth mine. On 4 September the bill reached the presidency, and on 16 September it was sanctioned as Law 15.506/2026 and regulated the same day by Decree 13.118/2026 (Section 8.1).

Within seventy-two hours, Brazil articulated a doctrine of conversion, legislated an instrument for it, and lost shareholder control of the asset the doctrine most obviously concerns.

Yang and Cimini saw the third event coming. Writing before the closing, they observe that the announced purchase of the Goiás mine by a North American company, tied to a long foreign supply contract, exposed the absence of a Brazilian table with a mandate to hold a strategic opinion on the matter. Their diagnosis is institutional: what is missing is a forum. This paper argues that something prior is missing as well. Even a table with full authority would find that the Brazilian trade statistic has no cell in which to record what it decided.

This paper does not dispute the doctrine. It asks a question the doctrine leaves open: **by what measure would a government establish, in ten years, whether conversion occurred?**

The question is not rhetorical. A policy that cannot fail cannot be corrected. Brazilian discussion of critical minerals in 2026 is unusually consistent in its normative content — the country should not repeat the iron-ore cycle, should not remain an exporter of raw material, should add value domestically — and equally consistent in offering no indicator by which any of these outcomes could be verified or refuted. Fiscal credits, a governing council, a guarantee fund and export-linked value-added commitments are all created in the same legislative text; none of them is attached to an observable quantity.

This paper proposes such an indicator, applies it, and reports a result that limits the proposal itself.

The indicator rests on a distinction that is old in trade economics: some goods are sold at prices formed by an external reference for their contained element, and others at prices formed by their specification. Conversion, on this reading, means moving a flow from the first category to the second. The contribution here is not that distinction but its application at the level of a single country's customs lines, year by year, and — more importantly — the discovery that the application mostly fails.

Of eight NCM-8 lines covering Brazil's lithium, rare-earth, niobium and magnet trade, **one satisfies the conditions under which the test can be run** — and it returns a commodity reading. It is spodumene concentrate, and it returns the answer everyone already expects. The remaining lines fail, and they fail for three distinct reasons, only one of which a customs reform would repair.

That result is the paper. A state that has voted a conversion policy can currently run the measurement on one of its critical-mineral chains, and on that one the answer is no.

Section 2 establishes the absence of a denominator in the Brazilian doctrine. Section 3 positions the argument against three literatures that already occupy adjacent ground. Section 4 sets out the test. Section 5 describes data and method. Section 6 reports results and the runnability matrix. Section 7 examines the rare-earth case and a

channel through which conversion value can leave a country without appearing in its trade account. Section 8 draws policy implications, including from a Chilean precedent that refutes the most obvious of them. Section 9 states what the instrument cannot do.

2. A doctrine without a denominator

Brazilian critical-minerals policy has acquired, over roughly three years, a coherent vocabulary and no arithmetic.

The vocabulary is that of conversion: *beneficiamento*, *agregação de valor*, *verticalização*, the refusal of the primary-exporter role. It appears in the Senate's own legislative analysis of the bill, in the mining industry's position papers, in the technical notes of the federal science and technology agencies, and in the commentary that accompanied the September vote. It appears, in its most articulate form, in Yang and Cimini's account of conversion as a doctrine of statecraft rather than a target of industrial policy.

The arithmetic is absent in a specific and consequential way. Every formulation names an end state — minerals transformed at home, value retained, dependence reduced — and none specifies the quantity whose movement would constitute evidence that the end state is approaching or receding. This is not a matter of missing statistics. Brazil publishes monthly, disaggregated, freely downloadable trade data at eight-digit tariff-line resolution. The data exist. What is missing is a convention that would make a subset of them into a measurement of the thing the policy claims to be doing.

The gap is visible inside the doctrine's most articulate statement. Yang and Cimini's fifth act is a national critical-minerals policy conditioning access to reserves, financing and licences on domestic processing and value addition, without discriminating by the origin of capital. Their concluding caution is that the more protected the strategic core, the more demanding its accountability must be: public targets, audit, transparency, periodic review (Yang and Cimini 2026).

The requirement is stated. The instrument that would satisfy it is not, and this paper is an attempt to supply one.

The distinction matters because it determines what kind of failure is possible. A doctrine with an end state and no denominator can be judged only by whether investment occurs, plants are announced, and credits are disbursed. All three are observable, and none of them is conversion. A refinery that operates at low utilisation, exporting a product priced off the same reference as the ore it replaced, satisfies every observable criterion in the current policy while changing nothing about Brazil's position in the chain.

Alain Desrosières's formulation is useful here, provided it is used in its weak form. To quantify, he writes, is first to agree on an equivalence and then to measure (Desrosières 2008). The strong reading of this — that the act of counting produces the capacity to govern — does not survive the Brazilian case and is not claimed. Lithium carbonate has had its own tariff line, 2836.91.00, throughout the period examined here. The convention exists. The state has not used it, and the line is nearly empty. Counting did not produce conduct.

The weak reading does survive, and it is the argument of this paper. Without a convention of equivalence, there is nothing for a doctrine to correct.

The rare-earth compounds leaving Minaçu are declared under a residual heading that does not distinguish a mixed carbonate from a separated oxide. No convention exists at that point in the chain. Consequently no statement about rare-earth conversion — that it is progressing, that it has stalled, that a policy accelerated or delayed it — can be either confirmed or refuted from Brazilian public data. The doctrine is not wrong about rare earths. It is, on the present nomenclature, unfalsifiable about them.

3. Positioning

Three literatures occupy ground adjacent to this argument. Each supplies something the paper uses, and each has a limitation that defines what remains to be done.

3.1 Product classification: Rauch (1999)

The distinction between goods priced by external reference and goods priced by specification is not new. James Rauch's classification divides internationally traded commodities into three categories (Rauch 1999): those traded on organised exchanges, those not exchange-traded but carrying a benchmark price quoted in trade publications, and all others, which he terms differentiated. The scheme has been in continuous use for a quarter-century and is still, so far as this author is aware, the most widely applied classification of its kind at a disaggregated level.

Two consequences follow for this paper.

First, the intuitive formulation of conversion as "selling outside commodity markets" is Rauch's differentiated category, and any test built on it must acknowledge the antecedent.

Second, and more usefully, Rauch's middle category — reference-priced goods, whose benchmark is quoted in an industry publication rather than set by an exchange — already names the awkward case that this paper's data produce. Ferro-niobium has no futures market. Its reference price is an assessment, and the assessment reflects the pricing of a producer holding the overwhelming majority of world supply. A test that compares the exporter's realised unit value to that reference is comparing a price to itself. Rauch identified the category in 1999; this paper encounters it in 2026 as a measurement failure rather than a taxonomic one.

The limitation is granularity. Rauch classifies at four-digit SITC. At that resolution, rare-earth compounds are a single object. The distinction between a mixed carbonate and a separated oxide — which is the entire substance of the rare-earth conversion question — is invisible. Rauch's own instrument is subject to the aggregation blindness this

paper diagnoses. The move made here is to shift the unit of analysis from the product to the flow: a given country's exports under a given code in a given year, assessed on their own behaviour rather than assigned by category.

3.2 Measuring downstreaming: the Indonesian literature

There is by now a substantial body of work evaluating whether export restrictions and processing mandates actually produce domestic value. Indonesia's nickel policy is its principal case. World Bank research measures the domestic value-added ratio of exports using firm-level customs transaction data (Kee and Xie 2025). Work published in *Mineral Economics* in 2025 applies intervention analysis with placebo tests and finds that the twenty-six-fold rise in processed nickel export value was driven predominantly by exogenous growth in electric-vehicle demand rather than by a discrete policy-induced shift (Soamole 2025). Input–output analysis published in *Economies* in 2026 maps the structural transformation of the domestic nickel ecosystem before and after the ban (Samputra and Djatmiko 2026).

This literature answers the question this paper asks. It is the standard, and it must be named as such.

It is also, for Brazilian purposes, unavailable. Domestic value added in exports is estimated from firm-level customs transactions. Brazil's public Comex Stat is aggregated by tariff line and destination country. No firm identifier is published. Any analyst outside the tax administration — and, on present arrangements, quite possibly the interministerial council the new law creates — cannot compute the standard measure.

The instrument proposed here is therefore a declared second-best: a test built on the data a public actor actually possesses. Its principal finding is that even the second-best fails on seven of eight lines. That is a stronger claim than it would be if the first-best were available and merely unused.

The Indonesian literature supplies one further thing directly. The finding that an apparent policy success was largely an exogenous demand trend is the methodological precedent for separating cycle from structure. It applies here in reverse: the collapse of Brazilian spodumene export value in 2025 is a price event, and reading it as a policy failure would be the same error in the opposite direction.

3.3 Evaluating industrial policy: Juhász, Lane and Rodrik (2024)

The current state of the art on measuring industrial policy is the survey by Réka Juhász, Nathan Lane and Dani Rodrik in the *Annual Review of Economics*, organised explicitly around measurement, causal inference and economic structure (Juhász, Lane, and Rodrik 2024).

One result from that literature is load-bearing here. Rodrik's earlier observation, restated in the survey, is that observational data alone cannot distinguish a technocratic government correcting a market failure from a rent-seeking government captured by interests (Rodrik 2009, restated in Juhász, Lane, and Rodrik 2024). The two produce the same observable trace.

This paper's identification problem is an instance of that result rather than a peculiarity of niobium. A flow whose unit value does not track a commodity reference may be a specified industrial product; it may equally be the output of a producer with sufficient market power to set its own price, or a transfer between related parties. The trade statistic cannot separate them. Naming this in advance is not caution; it is the correct description of the instrument's reach.

3.4 What is not claimed

The observation that customs nomenclatures are too aggregated to track critical-mineral value chains is established. UNCTAD's own work calls for improved granularity of trade classifications through engagement with the World Customs Organization precisely to

capture processing stages (UNCTAD 2025; WCO 2022). The European Commission's Joint Research Centre has mapped Combined Nomenclature codes to critical raw materials in detail (European Commission, Joint Research Centre n.d.). The diagnosis is occupied.

The question this paper adds is the inverse one. Given that the codes are aggregated, **how much of a single country's critical-mineral trade remains measurable, and for what distinct reasons is the rest not?**

4. The test

4.1 The criterion

A flow converts if, over a defined horizon, its price ceases to be formed by the market value of its contained element and comes to be formed by its specification.

The formulation is deliberately narrow. It says nothing about employment, technology transfer, or domestic ownership. It captures one thing: whether the good being sold has become something a buyer purchases for its properties rather than for the metal it contains.

One consequence should be stated before the test is applied rather than discovered afterwards. A good can undergo substantial physical transformation and still be priced off its contained element. Concentration, roasting, leaching and even some chemical conversion can leave the price mechanism intact. What the test detects is a change in how the price is formed, not the presence or absence of industry. A line that fails it has not been shown to be unprocessed; it has been shown to be still sold as a commodity. The two are not the same claim, and the second is the only one made here.

Two routes operationalise it.

4.2 The price route

For a flow i in period t , let

- p = FOB value ÷ net weight, from Comex Stat;
- π = the market price of the contained element;
- θ = the average declared or conventional grade;
- $r = \theta \times \pi$, the commodity-equivalent value of the flow;
- $s = p - r$, the spread.

Estimate β as the coefficient of p on r over a rolling 36-month window.

The reading is a pair, not a single statistic:

	β	Mean spread
Commodity	≈ 1	≤ 0 (payability, treatment discount)
Conversion	clearly < 1	> 0 , persistently signed

A coefficient of variation across destinations is retained as a robustness check only; it confounds freight, Incoterm and lot size.

The price route requires π . Where π is a licensed assessment rather than an exchange settlement, and where the assessed producer is the exporter being tested, β is tautological. This is Rauch's reference-priced category presenting as a measurement failure.

4.3 The relational route

The second route uses no external price series. For each line, computed from export records only:

- **Concentration:** mean annual Herfindahl index of destination shares by weight.
- **Persistence:** median of the longest run of consecutive years with non-zero volume, taken across destinations.
- **Rotation:** mean year-on-year symmetric difference of the destination set, divided by its union.

- **Dispersion:** coefficient of variation of the monthly unit value.

An industrial offtake and a commodity sale leave different traces. A supply relationship with transformers is dispersed across many buyers, long-lived, low-churn, and priced stably because the price is contractual. A commodity sold to refiners is concentrated on whoever is refining, short-lived per relationship, high-churn, and priced volatily because the price is the market's.

Section 6 shows that this route discriminates more sharply than the price route on the same data, and that it does so without any licensed input.

It remains a heuristic classifier rather than an identified test. A genuine industrial relationship can be concentrated on a single long-standing buyer, and a commodity can be sold to a handful of persistent traders. The destination structure separates contractual from market behaviour; it does not by itself establish what the counterparty does. That is why the counterparty condition is separate, and why indeterminacy counts against.

4.4 The counterparty condition

Neither route establishes what kind of firm is buying. Comex Stat records the destination country, not the importer or its sector. This must be stated in the text of any application, because a criterion that is postulated but presented as computed invalidates everything computed alongside it.

Three proxies are available, of decreasing rigour:

- **Mirror declaration.** Does the partner country record the flow under a manufacturing-input heading? Available from Comtrade, with the usual six-to-eighteen-month lag.
- **Structure.** Persistence and concentration, as above. This separates contractual from market flows; it does not separate transformers from refiners.

- **Contract evidence.** Documented offtake relationships, external to the trade statistic and cited as such.

Flows are coded **I** (transformer established), **R** (refiner, separator or trader established), or **U** (indeterminate).

The rule is asymmetric: **a flow coded U cannot validate a conversion.** Indeterminacy counts against, never in favour. Without this rule the counterparty condition becomes a device for rescuing whichever case the analyst prefers.

4.5 The clock

A ten-year horizon requires a start date. The rule adopted here: the clock begins in the first year of sustained commercial export under a heading of Chapters 25 or 26.

This is not the year of discovery, of reserve declaration, or of policy announcement. For Brazilian lithium it dates the clock from the spodumene export surge of 2018–2019 rather than from decades of small-scale carbonate production. For rare earths it dates from 2024.

4.6 The four conditions, and two distinct questions

The conditions answer two questions that must not be run together.

The first is whether the test can be applied at all. A line is **runnable** when three conditions hold: **C0** — mass, at least 24 months above the noise threshold, below which there is no series but only shipments; **C1** — grade, θ declared or fixed by commercial convention; and **C2** — independent reference, π formed independently of the exporter under test. C0 and C1 condition the computation, C2 its interpretation.

The second is what the test returns. A runnable line **validates conversion** only when the reading is one of conversion — β clearly below one with a persistent premium, or the relational signature of contractual industrial supply — and the fourth condition holds: **C3** — counterparty, the flow coded I.

The distinction matters because the two failures are different. A line that is not runnable tells us nothing. A line that is runnable and does not validate tells us that conversion has not occurred, which is a result. Confusing the two would let an absence of data pass for a finding.

4.7 What splitting a code does and does not do

One clarification belongs in the statement of the test rather than in its defence. Creating subheadings under a residual rare-earth compounds line does not produce a separation plant, a magnet factory, or a domestic buyer. It produces the possibility that the absence of these things can be recorded, dated, and attributed.

The instrument is one of accountability, not of causation. Everything downstream of that — the plant, the contract, the capital — lies outside what any customs code can reach.

5. Data and method

5.1 Source

All trade figures derive from the Ministry of Development, Industry, Trade and Services, Comex Stat, raw database (MDIC 2026). Two series are used:

- the national series, disaggregated to NCM-8, destination country, month;
- the municipal series, which is disaggregated to SH4 only.

The distinction is not incidental. A widely circulated figure for Brazilian rare-earth exports to China originates in the municipal series and therefore cannot, by construction, identify the eight-digit line. Section 7 resolves it by crossing the two.

Both series were extracted twice: once with the mass threshold applied, for the indicators, and once without it, for the accounting.

Extraction date: 6 September 2026. Comex Stat is revised retroactively. Replication should report its own extraction date rather than assume these values persist.

5.2 Scope

Eight NCM-8 lines, exports and imports, monthly, 2016–2025:

NCM	Description	Stage
2530.90.10	Spodumene / lithium minerals	0
2530.90.30	Other minerals	0
2836.91.00	Lithium carbonate	1
2825.20.20	Lithium hydroxide	1
2846.90.10	Praseodymium oxide	1
2846.90.90	Rare-earth compounds, other	1
7202.93.00	Ferro-niobium	2
8505.11.00	Permanent magnets, of metal	3

Subheading 2846.90 carries four national lines: .10 praseodymium oxide, .20 chlorides of the other rare-earth metals, .30 gadopentetate dimeglumine, and .90 other. Lines .10 and .30 record no exports over the period. Line .20 records 54 tonnes to Mexico in 2020–2021, outside the scope defined here and below the mass threshold used in Section 4. Version 4 of this paper described .10 as a residual heading for rare-earth compounds excluding cerium; that description was wrong and is corrected here, with the consequences set out in Sections 6.4 and 6.5.

Lithium-ion accumulators (8507.60.00) are retained for reference only. No mineral reference price applies to them, and the balance-of-trade framing they invite is addressed in Section 6.

5.3 Threshold

Months with less than 10,000 kg are excluded before computing unit values. Below that mass, unit values are dominated by samples, qualification shipments and rounding, and carry no information about commercial terms. The threshold is a parameter and is reported as such; results should be checked against alternatives.

The threshold governs the indicators, not the accounting. Months below it are excluded before unit values are computed and before CO is assessed, which is the point of it. But an annual total assembled from thresholded months only would be a lower bound on the line, and a share computed against such a total would be an upper bound. Two series are therefore used throughout: a thresholded series for the price and relational indicators and for CO, and a second, unfiltered extraction for every annual total and every share reported here.

The distinction is not cosmetic. Under 8505.11.00 the thresholded series records 20,578 kg for 2025 and nothing at all for 2024; the unfiltered line carries 38,033 kg and 15,428 kg respectively. A first version of this paper reported the thresholded figures as line totals and was wrong to do so.

5.4 Two identifier problems

Comex Stat identifies the municipality of Minaçu, Goiás, as 5313087. The national statistical institute identifies it as 5213087. Filtering the municipal base on the second returns zero records. Two federal administrations, two identifiers, one municipality — a minor instance of the paper's subject.

Second: this extraction gives Brazilian spodumene exports for 2025 as 111,690 tonnes for US\$ 66.6 million. A figure sourced from a commercial data provider and circulating in trade press gives 125,636 tonnes for US\$ 66.5 million — the same value, an eleven percent higher tonnage. This paper uses Comex Stat throughout and reports the divergence rather than reconciling it.

5.5 What is not included

The price route requires reference series for spodumene concentrate, lithium carbonate equivalent, neodymium-praseodymium oxide and ferro-niobium. These are published under licence and are not redistributed with this paper. The replication script accepts a user-supplied file.

Consequently no β is estimated here. The price route is specified, its failure conditions are established analytically, and its estimation is left to any user holding the licences. The relational route, which requires nothing beyond the public data, is estimated in full.

That asymmetry is itself a result, and it is discussed in Section 8.

6. Results

6.1 Ferro-niobium: legible, and reference-priced

Ferro-niobium is the only Brazilian critical-mineral line with a continuous ten-year series: 120 useful months of 120.

	Volume	Value	Unit value
2024	91,989 t	US\$ 2.38 bn	25.87 US\$/kg
2025	98,628 t	US\$ 2.66 bn	26.95 US\$/kg

The relational profile is unambiguous:

	Value
Mean destination HHI	0.231
Mean destinations per year	24.1
Median relationship duration	4 years

	Value
Mean rotation	0.252
CV of monthly unit value, 2021–2025	0.066

Dispersed across two dozen buyers, persistent, low-churn, and priced with a coefficient of variation of six percent over five years including a period of considerable commodity volatility. This is the signature of contractual industrial supply.

The 2025 destination structure supports the reading, with one reservation that should be stated rather than left for a reviewer. The eight largest destinations were China (49.7 percent of weight, 26.98 US\$/kg), the Netherlands (16.9 percent, 27.68), the United States (8.0 percent, 27.11), Singapore (7.7 percent, 25.44), South Korea (7.3 percent, 24.38), Japan (5.1 percent, 30.89), India (1.3 percent, 23.54) and Canada (0.8 percent, 25.64). Unit values across eight destinations span 23.54 to 30.89 US\$/kg — a range of under thirty percent on a flow of 99,000 tonnes. That uniformity is itself evidence of administered rather than negotiated pricing, and it reinforces the C2 failure discussed below.

The reservation concerns the counterparty coding. The Netherlands takes nearly seventeen percent of the flow, and Rotterdam is an entrepôt as much as a steel-consuming destination. Coding this line I rests on contract evidence external to the trade statistic, not on the destination structure alone. A share of the flow may be trade rather than transformation, and the trade statistic cannot separate them.

And yet ferro-niobium fails C2. There is no niobium futures contract. The reference price is a producer list price or an industry assessment reflecting it, and the exporter under test accounts for the overwhelming majority of world supply. A regression of the exporter's realised price on that reference measures the internal coherence of a dominant supplier, not the presence of specification pricing.

The correct conclusion is narrower than the one the case invites. Ferro-niobium does not validate the price metric. It validates a weaker and more defensible proposition: **a Brazilian processing stage becomes legible when the nomenclature separates it**. Without heading 7202.93, ferro-niobium would be recorded somewhere in Chapter 26 and there would be nothing to observe. Brazil inherited that separation from the international nomenclature; it did not create it.

6.2 Spodumene: the one runnable line, and the answer already known

Spodumene concentrate has 85 useful months, the second-longest series in the sample.

	Volume	Value	Unit value
2024	345,493 t	US\$ 285.9 m	0.83 US\$/kg
2025	111,690 t	US\$ 66.6 m	0.60 US\$/kg

The relational profile is the inverse of niobium's:

	Value
Mean destination HHI	0.978
Mean destinations per year	5.2
Median relationship duration	1.5 years
Mean rotation	0.673
CV of monthly unit value, 2021–2025	0.846

Near-total concentration combined with high churn. The flow goes overwhelmingly to one destination and, within that, to counterparties

that change from year to year. Unit-value dispersion is thirteen times that of ferro-niobium.

This is a commodity sold to whoever is refining. The counterparty is coded R, and the line therefore does not validate conversion. It is the only line in the sample where that verdict can be reached at all, because it is the only one on which the test is runnable.

6.3 The midstream: the falsification is an empty line

The most frequently repeated proposition in Brazilian mineral commentary is that the country exports ore and imports batteries. The import side is real: Brazil imported 21,389 tonnes of lithium-ion accumulators for US\$ 623.7 million in 2024, and 25,549 tonnes for US\$ 609.8 million in 2025. But as an argument about conversion it is weak, because the counterfactual is absent. Brazil could refine every tonne of its spodumene domestically and import the same volume of cells, since cell imports are driven by vehicle and electronics demand rather than by where the salt was made.

What falsifies conversion in lithium is not the balance. It is that the midstream line is empty.

NCM	2024	2025	Useful months
2836.91.00 lithium carbonate	292 t / US\$ 4.55 m	95 t / US\$ 1.33 m	13
2825.20.20 lithium hydroxide	0	0	0

Thirteen useful months in ten years, and volume falling by two-thirds between the two most recent — from 292 to 95 tonnes. This is not a chain forming. The tariff line has existed throughout; the convention was available; nothing was recorded against it.

Note the sequence this creates. In August 2024 the national development bank approved R\$ 486.7 million, within a total investment of R\$ 492.4 million and drawn from the climate fund it administers, for a facility at Itinga, Minas Gerais. The stated objective is to raise capacity from 270,000 to 520,000 tonnes per year of lithium concentrate. The bank describes the works as a *unidade de beneficiamento de lítio* — a processing unit. The capacity metric is stage zero. The vocabulary of conversion is applied, in the lender's own announcement, to an expansion of the line the doctrine proposes to move away from (BNDES 2024).

6.4 The runnability matrix

NCM	Mo.	2024	2025	US\$/kg 24 → 25	Run	Why not
2530.90.10	85	345 kt / US\$ 286 m	112 kt / US\$ 66.6 m	0.83 → 0.60	yes	— (returns: not converted)
2530.90.30	17	0.4 t / US\$ 67	0.6 t / US\$ 348	0.16 → 0.60	no	nomenclature
2836.91.00	13	292 t / US\$ 4.55 m	95 t / US\$ 1.33 m	15.59 → 13.92	no	missing industry
2825.20.20	0	nil	nil	—	no	missing industry

NCM	Mo.	2024	2025	US\$/kg 24→25	Run	Why not
2846.90.10	0	nil	nil	—	no	missing industry
2846.90.90	5	60.3 t / US\$ 2.17 m	679.4 t / US\$ 11.97 m	35.91 → 17.62	no	nomenclature
7202.93.00	120	92 kt / US\$ 2.38 bn	99 kt / US\$ 2.66 bn	25.87 → 26.95	no	market structure
8505.11.00	9	15.4 t / US\$ 0.39 m	38.0 t / US\$ 1.22 m	25.03 → 32.06	no	nomenclature

Mo. counts months above the noise threshold and governs C0, which is satisfied at 24 or more. Run indicates whether the test is runnable. Volumes, values and unit values in this table are annual line totals taken from the **unfiltered** series; the threshold governs the indicators, not the accounting (Section 5.3). Unit values are national-line averages across all destinations: the 17.62 US\$/kg for 2846.90.90 is the whole line, while the 17.65 US\$/kg used in Section 7.3 is the China flow alone.

The test is runnable on **one line in eight**. That line, spodumene concentrate, returns a commodity reading and a counterparty coded R. **No line in the sample validates conversion.**

By the relational route alone, which requires no licensed series, indicators can be computed on two lines in eight — spodumene and ferro-niobium, the only ones with sufficient months and destinations. Ferro-niobium reads as contractual industrial supply, but its counterparty coding rests on evidence external to the trade statistic and its price route fails C2, so the reading is reported as legibility rather than as validated conversion.

Neither figure is comfortable. Both are defensible, and the second is only reached by declaring what the first cannot establish.

6.5 Three causes, one remedy

The seven failures are not the same failure.

Nomenclature (three lines). 2530.90.30, 2846.90.90 and 8505.11.00 fail because the heading does not identify the object. A mixed rare-earth carbonate and a separated dysprosium oxide are the same code. An AlNiCo magnet, a sintered samarium-cobalt magnet and a sintered neodymium-iron-boron magnet are the same code. Grade cannot be declared because the heading does not ask for it.

The failure is not one of absent granularity. Brazil has already made three national cuts inside 2846.90 — praseodymium oxide, chlorides of the other rare-earth metals, and a medical contrast agent — and none of them catches the mixed carbonate that actually leaves Minaçu. What is required is the extension of a distinction the administration already knows how to draw, not its invention. The one rare-earth line Brazil has cut precisely records no exports at all.

Missing industry (three lines). Lithium carbonate, lithium hydroxide and praseodymium oxide fail because there is nothing to measure. The codes are adequate; the flows do not exist. No nomenclature reform changes this.

Market structure (one line). Ferro-niobium fails because the reference price is the exporter's own. No nomenclature reform changes this either.

This distinction is the paper's principal limitation on its own recommendation. **Splitting customs codes corrects one family of blindness out of three.**

The magnet line illustrates both the case for the split and its limits. Heading 8505.11 covers permanent magnets of metal; ferrite and other non-metallic magnets fall under 8505.19. Within the metallic heading, the three commercially distinct families — AlNiCo, sintered samarium-cobalt and sintered neodymium-iron-boron — are separated by the United States tariff at ten digits, each with its own statistical line, and by the European Combined Nomenclature at eight, which isolates magnets containing neodymium, praseodymium, dysprosium or samarium from other metal magnets (European Commission 2025). The European integrated tariff descends further still, distinguishing shapes and industrial destinations. Two of Brazil's principal trading partners have made the cut, on two different logics. Brazil has made none at that heading: 8505.11.00 is undivided.

Aligning it would make the line legible. It would not, on present volumes, make it runnable: 38.0 tonnes in 2025 and 15.4 in 2024, spread over every month of the year and dozens of destinations, with one month in 2025 clearing the mass threshold and nine across the decade.

What the line does show, without any external reference, is the same self-contradiction found under the rare-earth heading. Consignments of a hundred kilograms or more left Brazil in 2025 at unit values ranging from 2.43 to 211.49 US\$/kg. The three largest consignments — 20,088 kg to Paraguay in September, then 6,000 and 3,000 kg to the Netherlands in October and April — carry 76.5 percent of the weight at between 12 and 15 US\$/kg, but only 32.7 percent of the value. Taken across the whole year the two destinations account for 78.6 percent of the weight and 36.0 percent of the value. The remainder is a few tonnes invoiced at several times that price. A tariff heading under which goods differ by a factor of eighty-seven in unit value within a single year is not identifying a product.

What any of it was, the statistic cannot say, and this paper does not speculate. That is the correct treatment of a line that does not identify its object, and it is why the recommendation in Section 8.3 is a nomenclature split rather than an inference about content.

Two indicators are therefore required rather than one:

- a **legibility rate**, counting lines where grade is declarable — moved immediately by an administrative act;
- a **testability rate**, counting lines satisfying all four conditions — moved only by industry, contracts and market structure, over years.

The gap between them is itself informative. It measures what a state has made itself able to count, against what there is to count.

7. One line, one mine, one destination

7.1 Resolving the figure

A figure of approximately 678 tonnes of Brazilian rare-earth exports to China in 2025 has circulated in press coverage attributed to trade ministry data. It originates in the municipal series, which is disaggregated to SH4 only and therefore cannot identify the tariff line.

Crossing the two series resolves it.

Source	Volume	Value
Municipal series: Minaçu (5313087) × SH4 2846 × China	677,908 kg	US\$ 11,962,522
National series: NCM 2846.90.90 × China	677,908 kg	US\$ 11,962,522

The two agree exactly. The flow is 2846.90.90. It is not 2530.90.30, whose entire 2025 line is 583 kg across all destinations, three orders

of magnitude below the flow in question, nor 2846.90.10 (praseodymium oxide), which records no exports at all.

The tariff line is therefore establishable. The grade is not. 2846.90.90 is a residual heading for rare-earth compounds. It does not distinguish a mixed carbonate from a separated oxide, and it records no rare-earth content. The product leaving Minaçu is a mixed rare-earth carbonate. The heading is in Chapter 28, the inorganic chemicals chapter, which is sometimes taken in Brazilian debate to indicate that the export is already chemical rather than mineral. The chapter placement certifies nothing about separation.

7.2 The proportion

Total Brazilian exports under 2846.90.90 in 2025 were 679,424 kg. Minaçu's shipments under SH4 2846 were 678,962 kg, of which 677,908 kg went to China. The national series and the municipal series agree on the China figure to the kilogram.

The remainders are small and worth naming, because a reader who subtracts should land where the data does. Within Minaçu, 1,054 kg went elsewhere: 51 kg to the United States at 33.92 US\$/kg, and 1,003 kg to Australia declared at a total value of US\$ 16, an implied 0.016 US\$/kg, which is a declaration anomaly rather than a transaction. Outside Minaçu, the whole of the rest of the national line is 462 kg — 460 kg to Mexico and two consignments of one kilogram.

At national level the line is effectively unanimous in both years: China takes 99.8 percent of the unfiltered 2025 flow, and everything under this heading that clears the mass threshold, in 2024 as in 2025, goes there.

One municipality accounts for 99.93 percent of the national line.

One municipality-destination pair accounts for 99.78 percent.

The national tariff line and a single mining operation are, statistically, the same object. There is no competing flow under this heading against which a third-party price could be anchored.

(The same municipality exported 167,280 tonnes of chrysotile asbestos under SH4 2524 in 2025, for US\$ 85.2 million. This is a separate industry and does not belong in the rare-earth discussion.)

7.2b The heading contradicts itself in a single year

One further observation is contained entirely within this tariff line and requires no external reference.

In 2025 Brazil exported 679,424 kg under 2846.90.90 at an average 17.62 US\$/kg, and imported 887,078 kg under the same heading at 1.99 US\$/kg. Both figures are from the unfiltered series; the inbound line was re-extracted on 18 September 2026, the export series on 6 September.

A ninefold difference in unit value between the two directions of the same code in the same year. The outbound flow is a mixed carbonate rich in magnetic and heavy rare earths; the inbound flow, at that price, is something else entirely – most plausibly cerium-bearing material, for which there is little market. Both are, for Brazilian customs purposes, the same product.

This is the clearest available demonstration that the heading does not identify an object. It requires no reference price, no grade assumption and no external source. The line refutes its own homogeneity.

7.3 The unit value

Between 2024 and 2025 the unit value of this line fell from 35.91 to 17.62 US\$/kg – a decline of 51 percent – while volume rose from 60.3 to 679.4 tonnes, a factor of eleven.

Both figures are national-line averages; China takes essentially the whole line in both years, so this is not a destination-mix effect. On the China flow alone the 2025 figure is 17.65 US\$/kg and the decline is the same.

It is a price effect, a grade effect, or a change of product. Brazilian public statistics cannot establish which. The heading records neither

rare-earth content nor separation state, and no reference series applies to an unspecified mixture.

One reservation must be stated with the observation. The 2024 base is 60.3 tonnes across four months with any flow at all. That is a qualification volume, not a series. The comparison is reported as an observation that cannot be interpreted, not as a trend. That distinction is the point rather than a weakness of it: an eleven-fold industrial ramp with the unit price halving, in a chain the state proposes to subsidise, and no public means of establishing whether the country sold the same product more cheaply, a different product, or the same product at a different grade.

7.4 The capital-account channel

On 3 September 2026 USA Rare Earth completed its merger with SVRE Holdings Ltd., the British Virgin Islands company controlling Serra Verde, operator of the Pela Ema mine at Minaçu. The merger agreement is dated 19 April 2026; the closing 8-K was filed on 4 September and reports the transaction as of 3 September (USA Rare Earth 2026a; 2026c).

Consideration was US\$ 300 million in cash and 126,849,307 USAR shares. The valuation depends entirely on which share price is applied: the registration statement implies US\$ 19.95, the close on 17 April, the last trading day before the agreement; the Nasdaq close on the day of closing was US\$ 17.69 (USA Rare Earth 2026b). The frequently quoted figure of approximately US\$ 2.8 billion is the first of these, not the second, and should not be used without its denominator.

The financing structure is US-anchored on both sides. Serra Verde carries a development-finance facility with a ceiling of US\$ 565 million; the retained portion stood at US\$ 425.0 million outstanding at 30 June 2026, and an incremental tranche drawn on 4 June was extinguished at closing when the lender's warrants converted. A fifteen-year offtake covering 100 percent of Phase 1 production, with

contractual price floors for neodymium, praseodymium, dysprosium and terbium, was announced with the agreement in April; the special-purpose vehicle purchasing that output completed a US\$ 1.55 billion capitalisation on 24 August, and that capitalisation was a condition of the merger closing (USA Rare Earth 2026d).

Three things follow that the trade statistic cannot register.

First, the sequence separates control from flow. The offtake had to be in force for the merger to close, but deliveries under it had not begun at the extraction date. Two dates coexist in the file and both belong here: the operator's August statement expects deliveries to the purchasing vehicle to start early in the fourth quarter of 2026, while the pro forma disclosures place commercial operations in 2027. Neither had occurred by 6 September. The chain was therefore capitalised, and its control transferred, before any converted physical flow could appear in Comex Stat. The closing satisfies the condition "control changed". It does not satisfy the condition "a finished good is sold outside commodity markets".

Second, the consideration is principally equity. The Brazilian asset was paid for mainly in claims on a foreign listed vehicle whose value derives from downstream operations located elsewhere — separation, metal-making and magnet production outside Brazil. The acquirer's own projection of Serra Verde earnings is stated on the basis of separated oxide sales, which is to say on the basis of a processing step that occurs after the product leaves the country and that Brazilian nomenclature does not record. If conversion of Brazilian rare earths occurs, its value is capitalised in foreign equity. It passes through the capital account and never appears in the trade account.

This is the sharpest form of the paper's argument. When a chain becomes intra-firm and cross-border, customs statistics are not merely too aggregated. They are the wrong instrument. A country can host, permit and part-finance a conversion and record the same trade line as before.

Third, a corollary that limits the instrument proposed here. Without a related-party flag in the trade data, the unit value on this line will cease to be a market price once intra-group deliveries begin, and any price-route estimate for Brazilian rare earths is stillborn. The single new case in the sample is the case the metric cannot reach. It should be noted that as of the extraction date this is a legal consequence of the change of control rather than an observed fact in the data: the 2025 shipments predate the closing, and physical deliveries to the offtake vehicle had not commenced.

8. Policy implications

8.1 What the legislation can and cannot do

The bill was approved by the Senate on 2 September and sanctioned, without vetoes, on 16 September 2026 as Law 15.506/2026; Decree 13.118/2026, published the same day, regulates the council, which must be formally installed within ninety days.

The statute creates an interministerial council with authority over changes of control, international offtakes and mineral titles; a fiscal-credit programme capped at one billion reais a year from 2030 to 2034, at up to twenty percent of eligible spending, restricted to projects habilitated by the council; a guarantee fund; a research and development contribution assessed on gross revenue net of taxes; a public project registry; a traceability system; and an authorisation for the executive to attach value-added commitments to exports and to require, of exported critical minerals, the final beneficiary, the ownership chain, the degree of processing and the mineralogical composition (art. 8, III).

That last provision matters for the argument of this paper: the information whose absence is diagnosed in Sections 6 and 7 is now within the executive's power to require. Floor amendments sought to strike the export link and the words *final beneficiary* from it, to

suppress the provision in full, to replace homologation with registration and monitoring, and to confine traceability to the first link of the chain; none was accepted. The requirement survived the vote; the regulation that would make it bite has not been issued.

What the statute does not do is define the quantity against which conversion would be established. Value addition is to be defined "according to criteria set in regulation" (art. 2, XIX); the credit is to be proportional to value addition, with the minimum processing tiers for each band left to regulation (art. 18, §12); traceability is subject to a reasonable adaptation period, with no date (art. 44, §9); and the list of substances is fixed by the council, on a reasoned proposal from the Ministry of Mines and Energy.

Two further facts frame that delegation. Complementary Law 235/2026, of 27 August, disapplied for the 2026 financial year both article 14-A of the fiscal responsibility law and articles 140 to 149 of the budget guidelines law in respect of proposals instituting tax benefits for this policy — the provisions that would have required an estimate of beneficiaries and quantifiable performance targets. And article 46 of the law retains a five-year sunset citing the first item of that guidelines provision — duration, not performance — although the complementary law had already set articles 140 to 149 aside for this class of bills.

The council is a filter on ownership, not a measure of transformation, and it did not exist on 3 September, when the acquisition described in Section 7 closed.

8.2 The precedent that refutes the obvious remedy

The intuitive instrument — require exporters to offer product domestically at preferential prices to qualified transformers — has been tried in the region and has not produced a plant.

Chile's development corporation obliged Albemarle and SQM Salar, under contracts of 2016–17 and January 2018, to make available up to 25 percent of their lithium production at a preferential price to

productores especializados selected by tender, in exchange for expanded extraction quotas. The stated purpose was to install value-added lithium manufacturing in Chile.

The mechanism has been re-tendered four times in eight years.

The first round opened in April 2017 and drew twelve bids. Three were selected in March 2018: Molymet, a Samsung–POSCO consortium, and Sichuan Fulin. POSCO withdrew in June 2019, having required hydroxide where the available quota was carbonate; Molymet and Sichuan Fulin followed in July. The decisive fact is earlier than any of the withdrawals. The three winning projects together called for 28,496 tonnes of lithium against roughly 16,000 tonnes earmarked for them, and an internal corporation memorandum dated 9 March 2018 — the week of the award — recorded that it would be possible to satisfy Fulin and Molymet on the one hand, or Posco-Samsung on the other, but not all three (Reuters 2019). The mechanism was oversubscribed by nearly a factor of two at the moment it was awarded, and the awarding body knew it. Whatever the pricing formula did afterwards, the quota could not have supplied what had been promised. A second round selected the Chilean nanotechnology firm Nanotec in June 2020 (CORFO 2020), for a project of roughly one million dollars — an order of magnitude that indicates the difficulty of attracting the intended scale. A third round selected two Chinese-backed projects. BYD Chile was chosen in April 2023 for a 50,000-tonne LFP cathode plant with a minimum investment of US\$ 290 million (CORFO 2023). Yongqing Technology — a Wenzhou company 92 percent owned by Tsingshan Holding Group, and its new-energy arm — followed in October 2023, with a US\$ 233 million cathode plant of 120,000 tonnes a year, taking 11,244 tonnes a year of battery-grade carbonate at preferential price to 2030, half the quota then available. Both were due to be operating by May 2025. Neither plant was ever built. Both withdrew that month, and BYD's status was revoked in January 2026. A fourth call was announced in May 2024, this time against the Albemarle quota running to 2043.

The two failures have different causes, and both are instructive. BYD's withdrawal is attributed by the corporation to global market conditions affecting its investment decisions. But the record is more specific than a price story. Through 2025 the company asked either to extend its specialised-producer status beyond 2030 or to fix an alternative to the preferential-price calculation mechanism, in order to secure the viability of the business; the corporation replied that neither was contemplated in the tender rules. Its status was then withdrawn on 26 January 2026 for failure to begin operations by the deadline the rules set, 31 December 2025 (Dulanto 2026).

Yongqing's cause is not that. The corporation records that the project lapsed because **the company did not constitute in Chile the entity that was to carry it out**, in the terms the tender rules required. Constituting a local company was simultaneously a condition of retaining specialised-producer status and a ground for losing it without indemnity. Yongqing asked to be substituted by another Tsingshan entity; the request was refused because that entity had not participated in the proposal and had never been evaluated.

Two lessons follow, and both are developed in Section 8.3. From BYD: a published formula locks in both directions, and the Chilean rules contained no mechanism to revisit the preferential price when the reference had fallen by eighty percent. From Yongqing: a counterparty that exists only as a selected proposal is not a counterparty.

The claim that should be made is not a count of awards, which is disputable, but the outcome: **no value-added lithium plant of the scale the mechanism was designed to produce has entered operation in Chile under it**, eight years and four tenders after it began.

One figure on the corporation's own lithium page states the difficulty more economically than any narrative. Of the lithium hydroxide made available at preferential price under the SQM quota — 4,200 tonnes a year — the assigned monthly volume is zero (CORFO n.d.). Hydroxide is the product POSCO required in 2019 and could not obtain. It has since been offered, and no one has taken it. The same page, last

revised in May 2025 and therefore predating the revocation of January 2026, records that of the two producers selected in the round closed on 31 December 2024, one status remained in force; it does not name which.

The causes are transposable. The obligation ran to making product available at a preferential price, not to a firm sale of a firm volume. A counterparty that exists only as a selected proposal is not a counterparty: the Chilean rules required local incorporation and the largest applicant simply did not do it. The price formula — for Albemarle, a band set on the lowest-priced twenty percent of sales to unrelated third parties over the preceding six months, or an alternative schedule agreed with the miner — was litigated for years and then overtaken by a price collapse. The product was named imprecisely: the quota was carbonate where the intended plant required hydroxide. And the arrangement conferred an administrative status, revocable, rather than a supply contract.

Brazil has not copied this instrument. It has dispersed it: an export-linked value-added commitment with no formula, a council with no statutory deadline for its decisions, a fiscal credit with no tariff-line nexus, and a development bank financing stage zero. The risk is not repeating Chile. It is arriving at the same zero with two fewer instruments — without a contractual clause and without an observable price.

8.3 Four acts

None of the following creates a plant.

One. Nomenclature. Further subheadings under 2846.90 distinguishing mixed compounds from separated oxides and salts of neodymium, dysprosium and terbium — on the model of the cut already made for praseodymium oxide — and alignment of the magnet line with the European separation by rare-earth content. This moves the legibility rate. On present volumes it moves the testability rate for no line. Both statements should be made.

Two. Conditionality on irreversible acts. A fiscal credit conditioned on a ten-year outcome cannot cause what it measures. The Chilean failure is a failure of conditioning on promises; conditioning on results is the same error displaced in time. The observable, datable and unfakeable events are: a final investment decision; a licence at the chemical stage; a firm offtake with a transformer already constituted in Brazil; and product qualification by a downstream manufacturer — a long, expensive and documented process that no unconverted oxide obtains, and which for that reason cannot be simulated. Release the credit on these. Use the trade line afterwards, to verify that the act was not cosmetic.

Three. Four locks on the export commitment. Sell, not offer. A published formula. A product named by tariff line — 2836.91.00 or 2825.20.20, whichever the plant actually buys. And a counterparty already constituted in Brazil, not a selected project: the largest of the Chilean awards lapsed for precisely this reason, the applicant never having incorporated the local entity its own bid required.

A published formula locks in both directions, and the second direction must be designed. It stops the state improvising; it also stops the counterparty renegotiating when the market moves. Chile's rules contained no mechanism to revisit the preferential price, and when the reference fell by eighty percent the counterparty asked for one, was refused because the tender rules did not contemplate it, and left. A Brazilian clause that publishes its formula must therefore publish, at the same time, the conditions under which the formula may be reopened. Five elements, all in the original instrument:

Trigger. An objective threshold on a named public series — the reference price outside a defined band for a defined number of consecutive months — not a discretionary standard such as material adverse change.

Window and frequency. Reopening may be invoked only within a stated window and at most once in a stated number of years, so that a review clause does not become permanent renegotiation.

Scope. Exactly which parameters may move — the discount, the floor — and which may not. Volume, tariff line, counterparty and delivery deadline are not price parameters. The Chilean counterparty asked to extend the duration of its status, which no price-review clause should ever reach.

Procedure and fallback. Who decides, on what timetable, and what happens on disagreement — expert determination is faster than arbitration — with an express obligation to continue performing while the review runs.

A measurement boundary. This is the element the Chilean model has no reason to contain and a Brazilian one must. A revised price that restores co-movement between the flow's unit value and the contained-element reference has returned the good to commodity pricing, and with it has dissolved the conversion the clause exists to produce. The permitted scope of any reopening should be bounded by the test in Section 4: the formula may be revised, but not into one that the test would read as a commodity.

The indexation trap. The obvious remedy for rigidity is to index the formula to a published reference so that it never needs reopening at all. That remedy defeats the instrument. A price fully indexed to the contained element produces, by construction, a unit value that tracks that element — the commodity reading of the test, with a coefficient of one and no premium. The more robust such a clause is to market movement, the less conversion it can produce. This is not a detail of drafting; it is the same tension the whole paper describes, arriving in the contract.

The resolution is to index against the right series. A domestic supply obligation for lithium carbonate indexed to lithium carbonate equivalent is a commodity contract with extra steps. The same obligation indexed to a published price for a named battery-grade specification, carrying the premium that specification commands, is not. **The reference series in the formula should sit at the stage the policy is trying to reach, not the stage it is trying to leave.** Where no

such series exists — which is the case for most of the products Brazilian policy names — that absence is itself information, and it belongs in the instrument's design rather than being discovered after signature.

Schedule extensions are a separate clause. A price review is not an exit, and a missed deadline is not a price problem. The Chilean counterparty asked to extend the duration of its preferential-price status beyond 2030, and was refused because the tender rules did not contemplate it. The refusal was correct; the absence of any published alternative was not.

The design turns on who caused the delay, and the Chilean record supplies the taxonomy. The corporation's account is that it worked with the ministry responsible for public land to identify sites in Antofagasta, that the process advanced adequately, and that in January 2025 the counterparty discarded the site it had itself selected. Delay caused by the granting authority — licensing, land, permits, grid connection — and delay caused by the counterparty are different objects and should not be handled by the same rule.

Three published triggers follow. Delay attributable to the state extends both the construction deadline and the benefit window, day for day and automatically, because otherwise the state's own failure destroys the economics it induced the counterparty to accept. Delay attributable to the counterparty extends neither. Force majeure, narrowly defined and enumerated, extends the deadline but not the benefit window, so that the loss is shared rather than transferred.

Two further elements make the clause auditable rather than discretionary. Cumulative extension is capped, or the deadline is fictional. And the asymmetry of silence follows the rule used elsewhere in this paper: where the claim is that the state caused the delay, an authority that does not answer within the published period is taken to have granted it; where the claim is anything else, silence is a refusal. Every claimed delay, its attributed cause and its outcome should be entered in a public register. An instrument whose

exceptions are unpublished is not a rule with exceptions. It is discretion with a rule attached.

Four. A related-party flag. Whether an export is invoiced to an entity related to the exporter, recorded in the trade data and available to the authorities that homologate offtakes. Without this, the Minaçu unit value is not a price. The flag is not a precaution; after 3 September it is the only price anchor that remains for Brazilian rare earths.

8.4 Auditing is not steering

A doctrine that cannot be audited cannot be corrected. That is this paper's claim, and it is the condition Yang and Cimini themselves set when they require public targets and audit of any protected strategic core.

An auditable doctrine is not thereby a steered one. The customs line records in 2035 what a decision of 2026 produced; it does not turn the wheel. Steering conditions irreversible acts and uses the trade line afterwards. Where the chain is intra-firm, the relevant Brazilian instrument ceases to be the customs bureau: it is the tax administration, whose transfer-pricing regime reaches transactions between related parties, and the authorisation, which can require physical flows as the price of approval.

The distinction matters because conflating the two is the same nominalism this paper argues against. A denominator makes failure visible. It does not make success occur.

9. Limits

Two points are not a trend. The rare-earth unit-value observation rests on a 2024 base of 60.3 tonnes across four months with any flow at all. It is reported as uninterpretable, which is its value, and not as evidence of decline.

No β is estimated. The price route is specified and its failure conditions established analytically. Estimation requires licensed series not redistributed here. Readers holding them should estimate it and report against the specification in Section 4.

The relational route does not identify counterparties. It separates contractual from market flows. It does not distinguish a transformer from a refiner. That distinction rests on evidence external to the trade statistic and is coded accordingly, with indeterminacy counting against.

Market power and specification remain confounded. This is the general result restated in Section 3.3, not a defect peculiar to this application. It is why the test has a counterparty condition and why ferro-niobium is reported as legible rather than as converted.

Intra-firm flows defeat the instrument. Once a line is entirely related-party, its declared unit value is not a price. This is the condition of the single case in the sample that is genuinely new.

The threshold is a choice. Ten thousand kilograms per month excludes qualification shipments and rounding noise. It also excludes early-stage flows that a different analyst might wish to observe. Results should be checked against alternatives; the parameter is exposed in the replication code.

One country, one period, eight lines. Nothing here establishes that the coverage rate found for Brazil generalises. The method is transposable; the result is not.

10. Conclusion

Brazil has voted a conversion policy and can currently run the measurement on one of its eight critical-mineral trade lines. That line is spodumene concentrate, and it returns the answer the country's own commentary already assumes. No line validates conversion.

Of the seven lines on which the test cannot be run, three are blind because the nomenclature does not identify the object, three because the industry does not exist, and one because the reference price belongs to the exporter. A customs reform addresses the first family. It leaves the other two untouched, and this should be said by whoever proposes it rather than discovered by whoever evaluates it.

The rare-earth line acquired mass in 2025 and did not thereby become runnable. It became visibly blind: eleven times the volume, half the unit price, one municipality, one destination, and no public means of establishing what changed. The day after the Senate approved a policy whose purpose is to retain that transformation, the asset changed hands, and its value was capitalised in foreign equity that no Brazilian trade statistic will ever record.

A doctrine that cannot fail cannot be corrected. The niobium line shows the answer can be the customs bureau, once the identification problem is admitted. The lithium line shows the answer is already no. The rare-earth line shows what happens when the question cannot be asked at all.

Appendix A – Variables

Variable	Definition	Source
p	FOB value ÷ net weight, US\$/kg	Comex Stat
π	Market price of contained element	licensed, not included
θ	Average declared or conventional grade	commercial specification
r	$\theta \times \pi$	derived
s	$p - r$	derived

Variable	Definition	Source
β	OLS coefficient of p on r , 36-month rolling	derived
HHI	Mean annual Herfindahl of destination shares by kg	Comex Stat
Duration	Median longest run of consecutive non-zero years per destination	Comex Stat
Rotation	Mean year-on-year symmetric difference of destination set $\div$ union	Comex Stat
CV	Coefficient of variation of monthly p	derived

Appendix B – Relational indicators, all lines, exports

NCM	HHI	Destinations/yr	Duration (yr)	Rotation
2530.90.10	0.978	5.2	1.5	0.673
2530.90.30	0.958	1.6	2.0	0.406
2825.20.20	1.000	1.0	1.0	1.000
2836.91.00	0.615	6.6	1.0	0.638
2846.90.90	0.903	2.4	1.0	0.542
7202.93.00	0.231	24.1	4.0	0.252
8505.11.00	0.361	49.9	2.0	0.484

NCM	HHI	Destinations/yr	Duration (yr)	Rotation
8507.60.00	0.197	36.5	2.0	0.473

2846.90.10 (praseodymium oxide) records no exports in the period.

These indicators are computed on the full export record, without the mass threshold applied elsewhere, because destination structure is informative even at small volumes. The figure of 49.9 destinations a year for 8505.11.00 is an average across years with non-zero flow and reflects many very small consignments; it is not comparable with the small number of consignments that clear the mass threshold, discussed in Section 6.5.

Appendix C — Resolution protocol for the municipal figure

1. Municipal series, filter `CO_MUN = 5313087`, `CO_PAIS = 160`, `CO_ANO = 2025`. Record volume and value by SH4.
2. National series, filter `NCM ∈ {2530.90.30; 2846.90.10; 2846.90.20; 2846.90.30; 2846.90.90}`, same country and year. The four lines under 2846.90 are tested together, since the municipal series resolves only to SH4 2846 and cannot distinguish them.
3. Compare. Agreement to within rounding identifies the line.
4. Where the two diverge, report the divergence. A rare-earth export whose tariff line cannot be established from public bases is the paper's thesis in its purest form.

Appendix D — Replication

Data and code accompany this paper. See `README.md` for provenance, column dictionary, threshold justification, and the reference series that are not redistributed.

Two extractions are deposited. The thresholded monthly series supports the price and relational indicators and the C0 assessment. A second, unfiltered extraction of 6 September 2026 supplies every annual total and every share reported in Sections 6 and 7:

`v2_totaux_annuels.csv` for the eight lines, and

`v2_28469090_2025_mes_pais.csv` and

`v2_85051100_2025_mes_pais.csv` for the two lines where the difference between the two series is material.

The NCM description of 2846.90.10 was corrected in version 5 at source, in the generator rather than in the exported tables; both `t1_matriz_C0.csv` and `t1_matriz_C0_2_.csv` should be regenerated from that single source. The README records the treatment of subheadings 2846.90.20 and 2846.90.30.

References

Bachelin Sena, Antoine. 2026. "Brazil-Argentina Critical Minerals Cooperation: Lessons from ABACC." *CEBRI-Journal*, August 27, 2026. <https://cebri.org/revista/en/artigo/285/brazil-argentina-critical-minerals-cooperation>.

Banco Nacional de Desenvolvimento Econômico e Social (BNDES). 2024. "BNDES aprova R\$ 486,7 milhões para Sigma Lithium beneficiar lítio de forma sustentável." Agência BNDES de Notícias, August 29, 2024.

BioBioChile. 2026. "Los cuestionamientos cruzados entre BYD y Corfo por la fallida planta de litio en el norte de Chile." January 26, 2026.

Brasil. 2000. *Lei Complementar nº 101, de 4 de maio de 2000*, art. 14-A, acrescentado pela *Lei Complementar nº 224, de 26 de dezembro de 2025*.

Brasil. 2025. *Lei nº 15.321, de 31 de dezembro de 2025*, art. 149. *Diário Oficial da União*, Seção 1, Suplemento D, 31 de dezembro de

2025.

Brasil. 2026a. *Decreto nº 13.118, de 16 de setembro de 2026. Diário Oficial da União*, Seção 1, Edição Extra, 16 de setembro de 2026.

Brasil. 2026b. *Lei Complementar nº 235, de 27 de agosto de 2026. Diário Oficial da União*, 28 de agosto de 2026.

Brasil. 2026c. *Lei nº 15.506, de 16 de setembro de 2026. Diário Oficial da União*, Seção 1, Edição Extra, 16 de setembro de 2026.

Brasil. Congresso Nacional. Senado Federal. 2024. *Projeto de Lei n.º 2.780, de 2024: institui a Política Nacional de Minerais Críticos e Estratégicos*. Brasília. Matéria 174060.

Cooperativa. 2025. "Corfo defendió Estrategia Nacional del Litio tras cancelación de proyectos de BYD y Tsingshan." May 7, 2025.

Corporación de Fomento de la Producción (CORFO). n.d. *Convocatoria: productores especializados de productos de litio de valor agregado*. Accessed September 6, 2026.

Corporación de Fomento de la Producción (CORFO). 2020. *Resolución (E) N.º 506: selección de Productor Especializado, Nanotec S.A.* Santiago: CORFO.

Corporación de Fomento de la Producción (CORFO). 2023. "BYD Chile es la primera seleccionada por Corfo en el Llamado a Productores Especializados de Litio." Press release, April 18, 2023.

Desrosières, Alain. 1993. *The Politics of Large Numbers: A History of Statistical Reasoning*. Cambridge, MA: Harvard University Press.

Desrosières, Alain. 2008. *L'argument statistique I: pour une sociologie historique de la quantification*. Paris: Presses des Mines.

Diario Financiero. 2025. "Corfo tras desistimientos de proyectos de BYD y Tsingshan: 'Han visto afectadas sus decisiones de inversión por las condiciones de mercado global'." May 8, 2025.

Dulanto, Blanca. 2026. "Corfo responde a BYD por desistimiento de planta de litio y le quita estatus de 'productor especializado'." *Diario Financiero*, January 27, 2026.

European Commission. 2025. *Combined Nomenclature and TARIC*, heading 8505 11. Brussels: Directorate-General for Taxation and Customs Union.

European Commission, Joint Research Centre. n.d. *Raw Materials Information System (RMIS)*. Accessed September 6, 2026.

Ex-Ante. 2026. "La caída de los dos proyectos chinos para dar valor agregado al litio en Chile." January 28, 2026.

Ibarra, Valeria, and Patricia Marchetti. 2025. "Duro revés: BYD y Tsingshan desisten proyectos de plantas de baterías y cátodos de litio en el norte de Chile." *Diario Financiero*, May 7, 2025.

Juhász, Réka, Nathan Lane, and Dani Rodrik. 2024. "The New Economics of Industrial Policy." *Annual Review of Economics* 16: 213–42. <https://doi.org/10.1146/annurev-economics-081023-024638>.

Kee, Hiau Looi, and Enze Xie. 2025. "Nickel, Steel and Cars: Export Ban and Domestic Value-Added in Indonesia." Policy Research Working Paper 11249. Washington, DC: World Bank. <https://doi.org/10.1596/1813-9450-11249>.

La Tercera. 2024a. "Corfo duplica la cuota de litio autorizada para Albemarle en el Salar de Atacama hasta 2043." May 16, 2024.

La Tercera. 2024b. "Nuevo llamado de Corfo a productores especializados entregará el 25% del litio que produzca Albemarle hasta 2043." May 17, 2024.

Ministério do Desenvolvimento, Indústria, Comércio e Serviços (MDIC). 2026. *Comex Stat: base de dados bruta, NCM e municípios (SH4)*. Extractions of September 6 and September 18, 2026.

Pivotes. 2024. "Corfo prepara la licitación de la preciada cuota de litio de Albemarle."

Rauch, James E. 1999. "Networks versus Markets in International Trade." *Journal of International Economics* 48 (1): 7–35.
[https://doi.org/10.1016/S0022-1996\(98\)00009-9](https://doi.org/10.1016/S0022-1996(98)00009-9).

Reuters. 2019. "How Lithium-Rich Chile Botched a Plan to Attract Battery Makers." July 17, 2019.

Rodrik, Dani. 2009. "Industrial Policy: Don't Ask Why, Ask How." *Middle East Development Journal* 1 (1): 1–29.
<https://doi.org/10.1142/S1793812009000024>. (Widely cited as 2008, the year of submission; the issue is dated 2009.)

Samputra, Palupi Lindiasari, and Hary Djatmiko. 2026. "Domestic Structural Transformation in a Critical Mineral Economy: A Multisectoral Assessment of Indonesia's Nickel Downstreaming Strategy." *Economies* 14 (4): 133.
<https://doi.org/10.3390/economies14040133>.

Soamole, Syamsul Bachri. 2025. "Indonesia's Nickel Downstreaming in the Geopolitics of the Global EV Battery Industry: A Political Economy and Scenario Modeling Analysis." *Mineral Economics*, December 23, 2025. <https://doi.org/10.1007/s13563-025-00588-6>.

United Nations Conference on Trade and Development (UNCTAD). 2025. *SDG Pulse 2025*. Geneva: UNCTAD.

USA Rare Earth, Inc. 2026a. Current Report on Form 8-K (completion of merger with SVRE Holdings Ltd.). Date of report September 3, 2026; filed September 4, 2026; File No. 001-41711. SEC Accession No. 0001213900-26-097399. Washington, DC: U.S. Securities and Exchange Commission.

USA Rare Earth, Inc. 2026b. Registration Statement on Form S-3 (resale of 126,476,950 shares). File No. 333-298777; filed September 4, 2026. SEC Accession No. 0001213900-26-097756.

USA Rare Earth, Inc. 2026c. Current Report on Form 8-K (Agreement and Plan of Merger). Date of report April 19, 2026; filed April 20, 2026. SEC Accession No. 0001213900-26-045706.

USA Rare Earth, Inc. 2026d. Current Report on Form 8-K (offtake and special-purpose vehicle capitalisation). Filed August 24, 2026. SEC Accession No. 0001213900-26-092810.

USA Rare Earth, Inc. 2026e. Definitive Proxy Statement on Schedule 14A (DEFM14A). Filed July 24, 2026. SEC Accession No. 0001213900-26-081091.

World Customs Organization. 2022. *Harmonized System Nomenclature*, headings 2530, 2825, 2836, 2846, 7202, 8505. Brussels: WCO.

Yang, Philip, and Fernanda Cimini. 2026. "Sources of Brazilian Conduct: Seeds of Power and the Doctrine of Conversion." *CEBRI-Journal*, Snapshots, August 31, 2026. ISSN 2764-7897. <https://cebri.org/revista/en/artigo/286/sources-of-brazilian-conduct>.